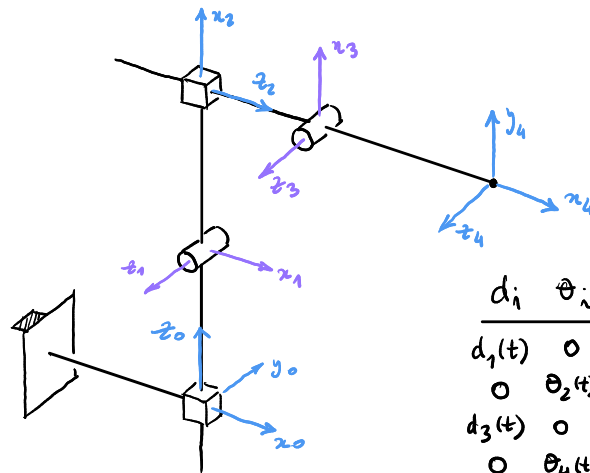
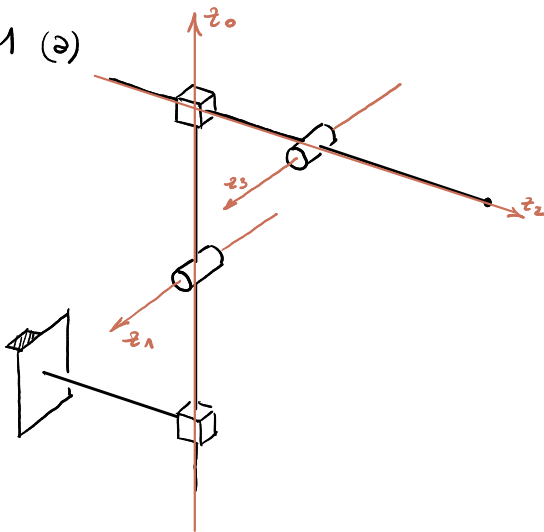
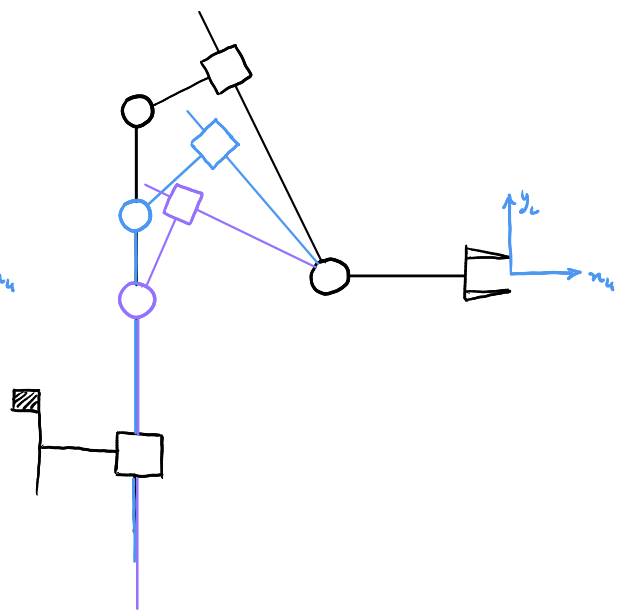
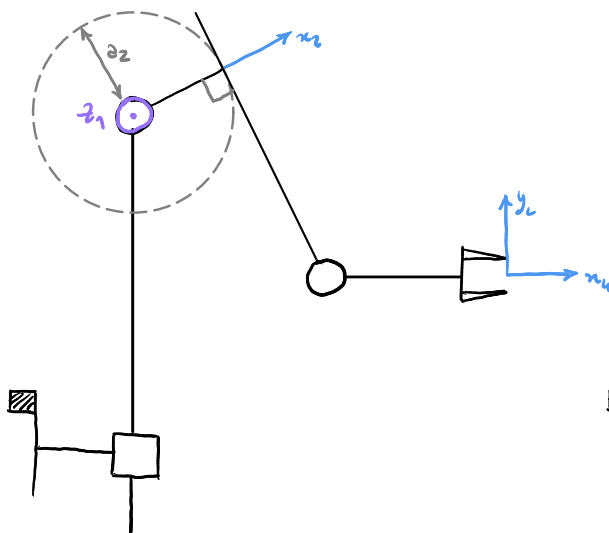


P1 (a)

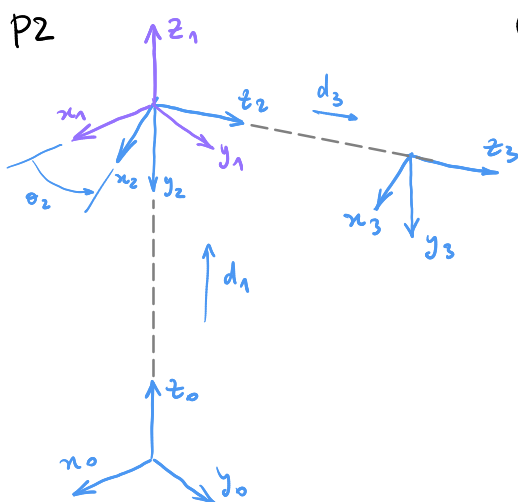


d_i	θ_i	a_i	α_i
$d_1(t)$	0	0	$\pi/2$
0	$\theta_2(t)$	a_2	$\pi/2$
$d_3(t)$	0	0	$-\pi/2$
0	$\theta_4(t)$	a_4	0

(b) To illustrate the nullspace motions, let Frame 4 be fixed in space.



P2



$$(a) \quad R_3 = \underbrace{R_1}_{=I} \underbrace{R_2^1}_{=I} R_3^2 = R_3^1 = (z_2^1 \ y_2^1 \ z_3^1) = \begin{pmatrix} c_2 & 0 & -s_2 \\ s_2 & 0 & c_2 \\ 0 & -1 & 0 \end{pmatrix}$$

$$p_3 = d_1 z_0 + d_3 z_3 = \begin{pmatrix} -d_3 s_2 \\ d_3 c_2 \\ d_1 \end{pmatrix}$$

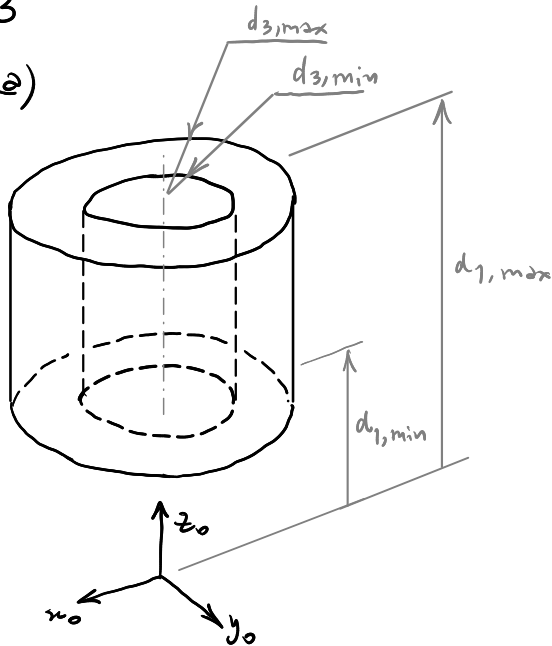
$$(b) \quad \mathcal{Y}_p = \frac{\partial p_3}{\partial q} = \begin{pmatrix} 0 & -d_3 c_2 & -s_2 \\ 0 & -d_3 s_2 & c_2 \\ 1 & 0 & 0 \end{pmatrix},$$

$$\mathcal{Y}_0 = (0 \ z_1 \ 0) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

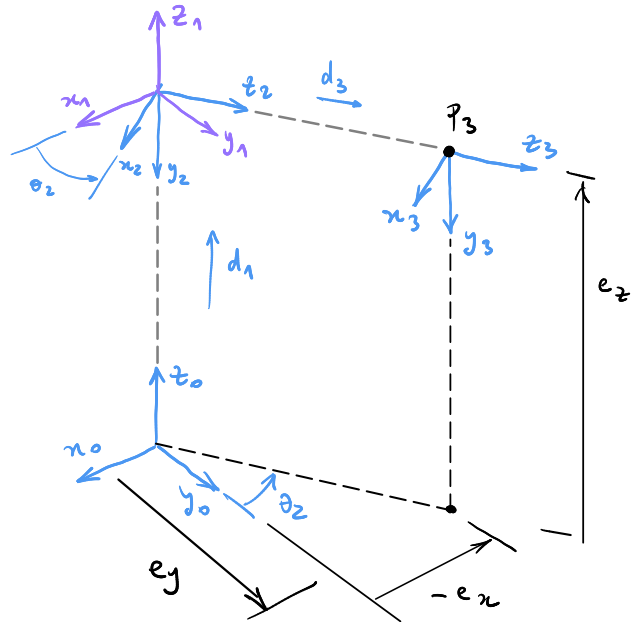
$$\mathcal{Y} = \begin{pmatrix} \mathcal{Y}_p \\ \mathcal{Y}_0 \end{pmatrix}.$$

P3

(a)



(b)

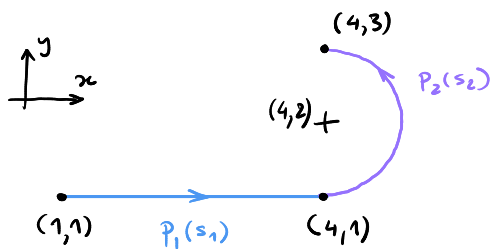


$$(p_3)_d \triangleq (e_x \ e_y \ e_z)^T,$$

$$d_3 = \sqrt{e_x^2 + e_y^2}, \quad d_1 = e_z, \quad \theta_2 = \arctan(-e_x, e_y).$$

P4

(a)



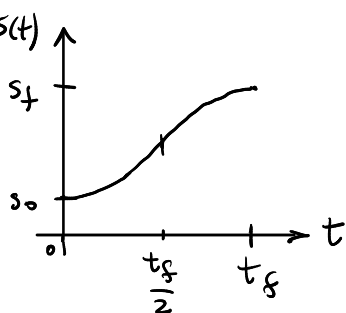
$$p_1(s_1) = \begin{pmatrix} 1+s_1 \\ 1 \end{pmatrix}, \quad 0 \leq s_1 \leq 3,$$

$$p_2(s_2) = \begin{pmatrix} 4 + \sin(s_2) \\ 2 - \cos(s_2) \end{pmatrix}, \quad 0 \leq s_2 \leq \pi,$$

$$s_1 = s, \quad s_2 = s - 3.$$

$$p(s) = \begin{cases} \begin{pmatrix} 1+s \\ 1 \end{pmatrix}, & 0 \leq s < 3 \\ \begin{pmatrix} 4 + \sin(s-3) \\ 2 - \cos(s-3) \end{pmatrix}, & 3 \leq s \leq 3+\pi \end{cases}.$$

(b)



$$\dot{s}(t) = \frac{6(s_f - s_0)}{t_f^2} t - \frac{6}{t_f^3} (s_f - s_0) t^2$$

$$\dot{s}_{\max} = \dot{s}(t_f/2) = \frac{6}{t_f^2} \Delta s \frac{t_f}{2} - \frac{6}{t_f^3} \Delta s \frac{t_f^2}{4}$$

$$\dot{s}_{\max} = \frac{6 \Delta s - 3 \Delta s}{2 t_f} = \frac{3(3+\pi)}{2 t_f}$$

$$t_f = \frac{3(3+\pi)}{2 \dot{s}_{\max}}.$$